SOURJYA BASU

FIRMWARE DEVELOPER | IoT | FREE-RTOS | COMPUTER NETWORKING

Intrigued by tinkering, learning and exploring new horizons…….

PROJECTS

PCD8544 FRAMEBUFFER DISPLAY DRIVER STACK FOR ESP32 (ESP-IDF)

- Built a **from-scratch custom LCD driver stack** for the Nokia 5110 LCD Module (**PCD8544 controller**) on Espressif Systems ESP32 using **ESP-IDF**, implementing **GPIO bit-banged serial transport**, **CE/DC/RST sequencing**, and **PCD8544** 48×84 framebuffer page writes
- Implemented a framebuffer flush pipeline with **scanline-to-8px-page conversion** and **full-frame serialized transfer into display controller RAM**.
- Developed an **integer-only raster engine** (Bresenham line + midpoint circle) with incremental decision logic and linked-list based **coordinate staging** for **deferred framebuffer rendering**.
- Built an 8×8 bitmap font renderer with **ASCII mapping,** and **direct framebuffer glyph composition.**
- Designed a **text layout + marquee system** supporting incremental framebuffer shifting for scrolling output.
- Created a **graphics driver API** for pixel, line, rectangle, triangle, circle and **batch framebuffer composition before hardware flush**.
- Integrated FreeRTOS for **timed execution and deterministic display refresh scheduling.**

CLOUDVPN+

Final Year Project 2025-2026

- Designed and implemented a **custom TCP tunnelling and multiplexing protocol** to route multiple logical streams over a single persistent connection, reducing connection overhead and improving efficiency
- Built from scratch using **raw sockets** for fine-grained control over protocol design, data flow, and connection management
- Developed **custom binary framing and stream identification** for packet routing and isolation
- Implemented **full-duplex communication** between client, server, and dynamic endpoints
- Designed a **multithreaded architecture** using producer–consumer queues for concurrent data transport
- Handled **TCP stream fragmentation and reassembly** for reliable data transmission
- Integrated **browser-based traffic routing** via PAC scripts with local client-server architecture.

CONTACT

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Vadodara , Gujarat

(Open to Relocation)

SKILLS

- Firmware Development
- FreeRTOS
- ESP-IDF
- Computer Networks
- Protocol Design
- MicroPython
- Embedded C
- Embedded Linux

CERTIFICATIONS

- CISCO Certified Network Associate
- NPTEL: Computer Networks and Internet Protocol

EDUCATION

PARUL UNIVERSITY, 2023-2027

B.Tech in Computer Science and Engineering

COMMUNICATION SKILLS

- Public Speaking
- Event Hosting & Anchoring
- Technical Presentation